

INDIAN STATISTICAL INSTITUTE

Linear Statistical Models

(B.Stat. 3rd year)

Semester 1, 2026-27

Homework 1

Due: August 19, 2026

1. Suppose $X \sim N_n(\mu, \Sigma)$. Show that, for a symmetric $n \times n$ matrix A ,
 - (a) $\mathbb{E}(X^T A X) = \text{tr}(A \Sigma) + \mu^T A \mu$.
 - (b) $\text{Cov}(X, X^T A X) = 2 \Sigma A \mu$.
 - (c) $\text{Var}(X^T A X) = 2 \text{tr}((A \Sigma)^2) + 4 \mu^T A \Sigma A \mu$.
2. Suppose $X \sim N_n(\mu, \Sigma)$ where Σ is positive definite. Let A and B be $n \times n$ symmetric matrices. Show that $X^T A X$ and $B X$ are independently distributed if and only if $B \Sigma A = 0$.
3. Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n \quad (1)$$

where ε_i 's are independent with $\mathbb{E}(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

Let $\boldsymbol{\beta} = (\beta_0, \beta_1)^T$, and let the least squares estimate of $\boldsymbol{\beta}$ based on observations $\{(x_i, Y_i)\}_{i=1}^n$ be denoted by $\hat{\boldsymbol{\beta}}^{(n)}$.

Now suppose we have an additional observation (x_{n+1}, Y_{n+1}) . Let $\hat{\boldsymbol{\beta}}^{(n+1)}$ denote the least squares estimate for $\boldsymbol{\beta}$ based on observations $\{(x_i, Y_i)\}_{i=1}^{n+1}$.

- (a) Show that $\hat{\boldsymbol{\beta}}^{(n+1)} - \hat{\boldsymbol{\beta}}^{(n)}$ can be expressed as

$$\hat{\boldsymbol{\beta}}^{(n+1)} - \hat{\boldsymbol{\beta}}^{(n)} = c_{n+1}(Y_{n+1} - \hat{Y}_{n+1}^{(n)})(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_{n+1}$$

where $\mathbf{x}_i = (1, x_i)^T$, $\mathbf{X}$ is the $n \times 2$ matrix with i -th row equal to $\mathbf{x}_i^T$, $c_{n+1} = (1 + \mathbf{x}_{n+1}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_{n+1})^{-1}$, and $\hat{Y}_{n+1}^{(n)}$ is the best linear unbiased prediction of Y_{n+1} based on observations $\{(x_i, Y_i)\}_{i=1}^n$.

- (b) Using the result in part (a), show that

$$Y_{n+1} - \hat{Y}_{n+1}^{(n)} = \frac{Y_{n+1} - \hat{Y}_{n+1}^{(n+1)}}{c_{n+1}}$$

where $\hat{Y}_{n+1}^{(n+1)}$ is the fitted value of Y_{n+1} based on least squares regression using data $\{(x_i, Y_i)\}_{i=1}^{n+1}$.

4. Consider the simple linear regression model (1). Assume further that errors ε_i are i.i.d. $N(0, \sigma^2)$. Suppose that we intend to test the hypothesis for **regression effect**, i.e., $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$.

- (a) Using the functional form of the least squares estimates $\hat{\beta}_1$ and $\hat{\sigma}^2 (= MSE)$, show that the t -statistic for testing H_0 , viz., $T = \hat{\beta}_1 / s(\hat{\beta}_1)$ has a $t_{(n-1)}$ distribution under H_0 .
- (b) Use the ANOVA decomposition: $SSTO = SSR + SSE$ associated with this regression model to claim that the F -statistic $F = MSR/MSE$ is of the form T^2 .
- (c) Argue that both tests will yield the same conclusion when performed at any level of significance.

5. Consider the simple linear regression model (1). Let $\hat{Y}_i$ denote the fitted value of Y_i using least squares regression based on data $\{(x_i, Y_i)\}_{i=1}^n$, and let $\hat{Y}_i^{(-i)}$ denote the fitted value of Y_i using least squares regression based on data $\{(x_j, Y_j)\}_{j \neq i}$ (i.e., excluding the i -th observation).

Let $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ denote the corresponding “hat matrix”, where $\mathbf{X}$ is the $n \times 2$ matrix with i -th row given by $\mathbf{x}_i^T = (1, x_i)$, for each i . Let h_{ii} denote the i -th diagonal element of the matrix $\mathbf{H}$. Using the results in Problem 3, or otherwise, show that

$$Y_i - \hat{Y}_i^{(-i)} = \frac{Y_i - \hat{Y}_i}{1 - h_{ii}} \quad \text{for } i = 1, \dots, n.$$

6. Consider the repametrization of the model

$$\mathcal{M}_1 : \quad \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I},$$

as

$$\mathcal{M}_2 : \quad \mathbf{Y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\eta}, \quad \mathbb{E}(\boldsymbol{\eta}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\eta}) = \sigma^2 \mathbf{I},$$

where $\mathbf{Z}$ has full column rank. Express the BLUEs of $\boldsymbol{\theta}$ and $\mathbf{X}\boldsymbol{\beta}$ in terms of one another.